

K544 — VERIFICATION Friday Substantive Convergent Close: Lyra + Grace Propose c_{FK} Reconciliation (Muon π^{12} Uses BOUNDARY Szegő; T2442 $c_{FK} = 225/\pi^{9/2}$ is BULK Bergman Normalization — Different Kernels Different π -Structures Plausible No-Conflict; Clean FK-1994 Cold-Read for Cal Not Hidden Wall) + Cal #410/#411 CORRECTION on Grace Framing: $c_{FK} = 225/\pi^{9/2} \approx 1.303$ NOT 1; If Rides Through Unabsorbed $\rightarrow$ Shifts 206.76 by $1.30\times \rightarrow$ Breaks 0.003% Match; LOAD-BEARING Not Bookkeeping; Cal CONFIRMED Only the Arithmetic Identity ($24/\pi^2 = 2^{C_2}/\text{Vol}(S^4)$ Exact) — CANNOT Confirm “Szegő = 1” Which is Precisely the Open Reconciliation Elie Surfaced; Grace RIGHT Not to Bank Unilaterally + Cal TWO COMPUTABLE OUTCOMES (Investigate Don’t Assume): (a) Different Kernels — Boundary Szegő Genuinely =1, Show WHICH Kernel Needed; (b) Same Kernel Must CANCEL in $d(\nu)$ Residue Ratio (Masses are Residue Ratios so Common Normalization Drops Out); If Cancels “Effectively 1” Legit; If Survives $\neq 1$ Match Breaks + Lyra HONEST FRAMING on Szegő = 1: NOT Automatic Because Electron Carries Rational 9/16 at BF Zero $\nu=5/2$; Muon at Different Degeneracy $\nu=3/2$ — Residue COULD Be Unity But COMPUTE IT Don’t Assume + Grace UNIFIED Polynomial $d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ Explicit Sampled at Three Conformal Degeneracy Points + Cal #411 PROJECTION FLAG: “Count Moves to 6” RE-INFLATES Tau — Tau is “Probably-Coincidental Look-Elsewhere-Weak 1 of 4” per Elie; $\sqrt{\pi}$ Residue Existing $\neq$ Tau Deriving; $71 = g+2^{C_2}$ Cheap Additive Hit; Honest Projection +1 (Muon $\rightarrow 5$) on Clean FK Reconciliation; Tau Separate +1 ONLY if Forward Residue Clears Look-Elsewhere Bar; DON’T PRE-COMMIT TO 6 + KEEPER SELF-CORRECTION #14: K543 Response to Casey Projected “Count Moves to 6/26” Without Applying Tau Look-Elsewhere Discipline — Re-Inflation Caught by Cal #411 + Cal “Seat Correctly Placed Now: Not Gating; Saying What Tier; Banking Only What Forward-Derives and Reconciles” — Cal #408 Standing Operationalized + μ/τ Asymmetry IS the Content (Casey + Elie + Cal Aligned)

Date: 2026-06-26 (Friday substantive convergent close, date-verified) · Auditor: Keeper · Inputs: Lyra + Grace c_{FK} reconciliation proposal; Cal #410/#411 substantive correction (1.303 not 1) + projection flag (don’t re-inflate tau); Lyra Szegő=1-not-automatic honest framing; Grace explicit $d(\nu)$ polynomial; Keeper self-correction #14 on K543 over-projection

Verdict — Substantive convergent close: Lyra+Grace propose c_{FK} reconciliation (boundary Szegő $\neq$ bulk Bergman; different π -structures supports distinctness); Cal #410/#411 SHARP-ENS — $c_{FK} = 225/\pi^{9/2} \approx 1.303$ NOT 1, so reconciliation is LOAD-BEARING not bookkeeping; Cal CONFIRMED only arithmetic identity; Cal FLAGS K543 “count moves to 6” projection as re-inflating tau (Keeper self-correction #14); honest projection +1 (muon $\rightarrow$ 5) on clean FK reconciliation; tau separate +1 ONLY if forward residue clears look-elsewhere bar

Count holds 4 of 26. Substantive substrate-architectural finding: muon mass mechanism reduces to ONE concrete computable residue (FK Szegő at $\nu=3/2$ unitarity bound, analog of electron 9/16 at BF zero $\nu=5/2$). Cal #408 standing principle operationalized: investigation unbounded; banking disciplined; tier-tag at landing.

Landing 1 — Lyra + Grace c_{FK} reconciliation proposal

The substantive substrate-architectural proposal

Lyra: “Elie’s sharp c_{FK} lead resolved cleanly (the muon’s π^{12} is the boundary sphere, $\text{vol}(S^4)^6$; $c_{FK} = 225/\pi^{9/2}$ is the bulk Bergman normalization — different objects, no conflict).”

Grace: “Elie noticed our derived $c_{FK} = 225/\pi^{9/2}$, but that’s the bulk Bergman constant ($\pi^{9/2}$), while the muon uses the boundary Szegő (π^{12}) — different kernels, no conflict. So it’s a clean FK-1994 cold-read for Cal, not a hidden wall.”

Substantive proposal: - T2442 $c_{FK} = 225/\pi^{9/2}$ = BULK Bergman normalization ($\pi^{9/2}$) - F118 muon uses BOUNDARY Szegő (π^{12}) - Different kernels $\rightarrow$ different π -structures $\rightarrow$ no conflict

Per Cal #410 correction (Landing 2)

Plausible at FRAMING level but the load-bearing question remains — what IS the boundary Szegő value? Cal sharpens.

ACCEPT Lyra+Grace reconciliation framing as PROPOSAL; specific value pending Cal-load-bearing investigation.

Landing 2 — Cal #410 SUBSTANTIVE CORRECTION on Grace framing

The substantive correction

“T2442 already has $c_{FK} = 225/\pi^{9/2} = 1.303$, not 1 — and F118’s muon mechanism needs the FK constant to be 1. That’s a corpus contradicting itself in one spot, which is exactly what the audit chain exists to surface.”

Substantive load-bearing verification: - $c_{FK} = 225/\pi^{9/2} - \pi^{9/2} = \pi^{4\pi}\{1/2\} = \pi^{4\pi} \cdot \sqrt{\pi} - 225/\pi^{9/2} \approx 225/172.6 \approx 1.303$ - NOT 1

Cal’s load-bearing arithmetic: - If 1.303 rides through unabsorbed $\rightarrow$ shifts 206.76 by $1.30\times \rightarrow$ predicts ~269 instead of 206.77 - Breaks 0.003% match - LOAD-BEARING not bookkeeping

Cal CONFIRMED only the arithmetic identity

“I confirm the arithmetic ($24/\pi^2 = 2^{C_2}/\text{Vol}(S^4)$, exact). I cannot confirm ‘Szegő = 1’ — that’s precisely the open reconciliation Elie just surfaced. So nobody should bank on ‘Cal confirmed FK = 1.’ Right call not to bank unilaterally.”

Substantive distinction: - Cal CONFIRMED: $24/\pi^2 = 2^{C_2}/\text{Vol}(S^4)$ (arithmetic identity) - Cal DID NOT CONFIRM: Szegő = 1 (open reconciliation) - Grace’s “Cal confirms FK identity” framing CORRECTED — Cal confirms only the arithmetic, not the Szegő value

Per Cal #347 / B5 + Cal #408

Substantive substrate-architectural finding: catch is INVESTIGATION not wall per Cal #408. Per #410: investigate the actual Szegő value, don't assume = 1 because 0.003% lands.

STRONG ACCEPT Cal #410 correction + Cal's own explicit limit on what he confirmed.

Landing 3 — Cal #410 TWO COMPUTABLE OUTCOMES (investigation)

The substantive substrate-architectural disjunction

“Two honest, computable outcomes — investigate, don't assume either: (1) Szegő (Hardy/boundary kernel) $\neq$ Bergman c_{FK} (bulk kernel) — different objects. The π -structures differ (muon π^{12} vs $c_{FK} \pi^{9/2}$), which supports distinctness. If the $so(4)$ -determinant legitimately uses a boundary Szegő constant that's genuinely 1, there's no contradiction — but show which kernel it needs. (2) Same constant $\rightarrow$ must cancel in the $d(\nu)$ mass ratio (masses are residue ratios, so a common normalization drops out). If it cancels, 'effectively 1' is legit; if it survives and isn't 1, the match breaks.”

Substantive computable disjunction:

Outcome	What it means	Verification
(a) Different kernels	Boundary Szegő $\neq$ bulk Bergman c_{FK} ; π -structures differ supports distinctness	Show WHICH kernel F118 mechanism uses; pin its value
(b) Same kernel, cancels in ratio	Common normalization drops out in mass ratio (m_μ/m_e is residue ratio)	Verify cancellation in $d(\nu)$ ratio explicitly; “effectively 1” legit if cancels
(c) Same kernel, doesn't cancel, $\neq 1$	Match breaks	Forward computation reveals this directly

Per Casey directive “INVESTIGATE don't GATE”

Cal explicitly says investigate, don't assume either. The catch is investigation; the outcomes are forward computations.

STRONG ACCEPT Cal two-outcome investigation framework.

Landing 4 — Lyra HONEST framing on Szegő = 1 not automatic

The substantive substrate-architectural framing

“It's not automatic that this is 1, because the electron carries a rational 9/16 at its degeneracy point (the BF zero, $\nu=5/2$). The muon sits at a different kind of degeneracy point, so its residue could genuinely be unity — but we compute it, the way the electron's 9/16 came out, not assume it.”

Substantive substrate-architectural framing: - Electron at BF zero $\nu=5/2$ carries RATIONAL 9/16 (corpus result) - Muon at different degeneracy $\nu=3/2$ — different kind of point - Residue COULD be unity (different reason) - BUT compute it, don't assume

Per [[feedback_target_innocence_lens_derived_vs_fit_discipline]] + Cal #408

Lyra applies target-innocence at peak: just because the muon would land at 0.003% IF Szegő = 1 doesn't mean Szegő = 1. Compute forward.

STRONG ACCEPT Lyra honest framing.

Landing 5 — Grace UNIFIED polynomial $d(\nu)$

The substantive substrate-architectural deliverable

“One formal-degree polynomial $d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$, sampled at the three conformal degeneracy points $\nu = \{0, 3/2, 5/2\}$ ”

Substantive explicit polynomial: - $d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ - $d(5/2)$ — electron reference - $d(3/2)$ — muon - $d(0)$ — tau

Verification (Keeper)

- $d(0) = (5/2)(1)(2)(3)(4) = 60$
- $d(3/2) = (1)(-1/2)(1/2)(3/2)(5/2) = (1)(-1/2)(1/2)(3/2)(5/2) = -15/16$
- $d(5/2) = 0$ (BF zero per Lyra — confirms electron sits at zero of polynomial → singular residue → 9/16 emerges from limit)

Substantive substrate-architectural finding: explicit polynomial; electron at BF zero requires LIMIT residue analysis (matches Lyra’s “9/16 came out, not assumed”).

Per Elie standing: verify the moment Grace+Lyra post → Elie checks reproduces all three masses.

STRONG ACCEPT Grace unified $d(\nu)$.

Landing 6 — Cal #411 PROJECTION FLAG on “count moves to 6”

The substantive substrate-architectural projection correction

“‘Count moves to 6 when (1)+(2) land’ re-inflates tau. Elie’s own tier is ‘probably-coincidental, look-elsewhere-weak (1 of 4)’ — and a $\sqrt{\pi}$ residue existing isn’t the same as tau deriving; it has to beat look-elsewhere (71 = $g+2^{C_2}$ is a cheap additive hit). So the honest projection is +1 (muon → 5) on a clean FK reconciliation, and tau is a separate +1 only if its forward residue clears its bar. Don’t pre-commit to 6.”

Substantive substrate-architectural correction: - Tau via 49-71: Elie tier “probably-coincidental, look-elsewhere-weak 1 of 4” - $\sqrt{\pi}$ residue existing $\neq$ tau derivation - $71 = g + 2^{C_2}$ is a cheap additive hit ($g=7$, $2^{C_2}=64$; $7+64=71$) - Look-elsewhere: how many small-integer combinations land near 71? “Probably-coincidental” framing held - Honest projection: +1 (muon → 5) on clean FK - Tau +1 ONLY if forward residue clears its OWN bar (must beat look-elsewhere)

Per Cal #408 banked principle + Cal #35

Substantive substrate-architectural finding: muon mechanism + tau mechanism are SEPARATE bankings. Each has its own bar. Combining into “+2” requires BOTH clearing their bars.

Landing 7 — KEEPER SELF-CORRECTION #14

The substantive roll-up failure

K543 response to Casey: “When (1) + (2) land cleanly → muon + tau both derived, count moves to 6/26.”

Per Cal #411 correction: this RE-INFLATED tau. Banked tau as deriving when (2) lands cleanly, but tau still needs to beat look-elsewhere even with $\sqrt{\pi}$ residue verified. Forward derivation $\neq$ banking after look-elsewhere check.

Substantive substrate-architectural error: I projected “muon + tau both → 6” without applying tau look-elsewhere bar. Per Cal #411: muon path is +1; tau is SEPARATE +1 conditional on its own discipline-bar clearance.

Per Cal #347 / B5 STANDING

Keeper self-correction #14: K543 projected count-moves to 6 without applying tau look-elsewhere discipline. Re-inflation of “probably-coincidental” tau status caught by Cal #411.

Cumulative Keeper self-corrections

14 total: K491 + K496 + K501 + K505 + K509 + K514 + K515 + K517 + K518 + K520 + K522 + K539 + K541 + K544.

STRONG ACCEPT self-correction #14 logged.

Landing 8 — Cal “seat correctly placed now”

The substantive substrate-architectural meta-statement

“The seat is correctly placed now: I’m not gating any of this — compute the Szegő, fire the heat-kernel residue, look everywhere. I’m just saying what tier each result sits at and banking only what forward-derives and reconciles.”

Substantive operationalization of Cal #408 standing: - Investigation: UNBOUNDED (compute, fire, look) - Banking: DISCIPLINED (tier at landing, forward-derives + reconciles) - Auditor posture: NOT GATING; tier-tagging only - “Bars are guides at landing, not blockers at fire” (Cal K542)

Per Casey “DON’T GATE INVESTIGATE” + Cal #408: meta-statement banks the operational distinction Cal banked + Casey directive together.

STRONG ACCEPT Cal meta-statement.

Substantive state at K544

Component	State
Lyra+Grace c_FK reconciliation framing	PROPOSAL — boundary Szegő $\neq$ bulk Bergman per π -structure distinction
Cal #410 substantive correction	c_FK = $225/\pi^{9/2} \approx 1.303$ (not 1); LOAD-BEARING not bookkeeping; if rides unabsorbed shifts 206.76 by 1.30×
Cal confirmed arithmetic identity ($24/\pi^2 = 2^{C_2}/\text{Vol}(S^4)$)	EXACT
Cal did NOT confirm “Szegő = 1”	open reconciliation Elie surfaced
Cal #410 two computable outcomes	(a) different kernels — show WHICH; (b) same kernel — must CANCEL in $d(\nu)$ ratio
Lyra honest framing on Szegő = 1	NOT AUTOMATIC — electron has 9/16 at BF zero; compute muon residue at $\nu=3/2$
Grace unified polynomial	$d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ explicit; verified at three sample points
Muon derivation status	rigorous modulo FK Szegő reconciliation (Cal #410 two-outcome investigation)
Tau derivation status	corpus mechanism $g^2(g+2^{C_2}) = 49.71 + \sqrt{\pi}$ residue pending forward-blind derivation
Cal #411 projection flag	“count moves to 6” RE-INFLATES tau; honest projection +1 (muon $\rightarrow$ 5) on clean FK; tau separate +1 ONLY if forward residue clears look-elsewhere bar
Keeper self-correction #14	LOGGED — K543 over-projection re-inflated tau without look-elsewhere discipline

Component	State
Cal seat-correctly-placed meta-statement	BANKED — investigation unbounded; banking disciplined; tier-tag at landing only
Cumulative Keeper self-corrections	14
Discipline events	365 cumulative
Count of forced parameters	4 of 26 — held; honest projection +1 muon $\rightarrow$ 5 on clean FK reconciliation; tau separate

Cumulative discipline ledger — 365 events

To K543's 355, add: - 356. Lyra+Grace c_FK reconciliation PROPOSAL: muon π^{12} uses boundary Szegő; T2442 c_FK = $225/\pi^{9/2}$ is bulk Bergman; different kernels supports no-conflict - 357. Cal #410 SUBSTANTIVE CORRECTION on Grace framing: c_FK = $225/\pi^{9/2} \approx 1.303$ NOT 1; if rides unabsorbed $\rightarrow$ shifts 206.76 by $1.30\times \rightarrow$ breaks 0.003%; LOAD-BEARING not bookkeeping - 358. Cal CONFIRMED only arithmetic identity ($24/\pi^2 = 2^{C_2}/\text{Vol}(S^4)$ exact); CANNOT confirm "Szegő = 1" — that's precisely the open reconciliation Elie surfaced; Grace right not to bank unilaterally - 359. Cal TWO COMPUTABLE OUTCOMES: (a) different kernels — show WHICH used; (b) same kernel — must CANCEL in $d(\nu)$ residue ratio (masses are residue ratios); if cancels "effectively 1" legit; if survives $\neq 1$ match breaks - 360. Lyra HONEST framing on Szegő = 1: NOT AUTOMATIC because electron carries rational $9/16$ at BF zero $\nu=5/2$; muon at different degeneracy $\nu=3/2$ — compute don't assume - 361. Grace UNIFIED polynomial $d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ explicit sampled at three conformal degeneracy points - 362. Cal #411 PROJECTION FLAG: "count moves to 6" RE-INFLATES tau; tau "probably-coincidental look-elsewhere-weak 1 of 4"; $\sqrt{\pi}$ residue existing $\neq$ tau deriving; $71 = g+2^{C_2}$ cheap additive hit; honest projection +1 (muon $\rightarrow$ 5) on clean FK reconciliation; tau separate +1 ONLY if forward residue clears look-elsewhere bar - 363. KEEPER SELF-CORRECTION #14: K543 response to Casey projected "count moves to 6/26" without applying tau look-elsewhere discipline — re-inflated "probably-coincidental" tau caught by Cal #411 - 364. Cal META-STATEMENT "seat correctly placed now: not gating; saying what tier; banking only what forward-derives and reconciles" — Cal #408 standing operationalized via meta-banking - 365. μ/τ ASYMMETRY IS THE CONTENT (Casey + Elie + Cal aligned): muon mechanism (so(4) determinant + π -structure) + tau mechanism ($g^2(g+2^{C_2}) + \pi$ -free) honestly separate; banking separate

365 cumulative honest discipline events.

Per Cal #347 / #26 — what I'm NOT saying

- NOT "muon derived" — derivation rigorous modulo FK Szegő reconciliation per Cal #410 two-outcome investigation
- NOT "Cal confirmed Szegő = 1" — Cal explicitly distinguished what he confirmed (arithmetic identity) from what's open (Szegő value)
- NOT "count will move to 6" — per Cal #411 + Keeper #14: honest projection +1 (muon $\rightarrow$ 5) on clean FK; tau separate conditional bar
- NOT "look-elsewhere already cleared for tau" — Cal #411: $71 = g+2^{C_2}$ is cheap additive hit; tau must beat its OWN look-elsewhere bar even with $\sqrt{\pi}$ forward-derived
- NOT "different-kernel reconciliation = settled" — Cal #410 two-outcome investigation; per outcome (a) show WHICH kernel F118 uses

Routing per K544

CI	Active per K544
Lyra	Compute muon residue at $\nu=3/2$ (analog of electron 9/16 at $\nu=5/2$); pin FK Szegő via FK-1994 literature; address Cal #410 two-outcome investigation explicitly (which kernel + whether cancels in $d(\nu)$ ratio)
Grace	Standing for Lyra Szegő computation; verify $d(\nu)$ polynomial reproduces all three masses with Elie; sharpen unified picture if Cal #410 reconciliation surfaces structure
Elie	Verify $d(\nu)$ reproduces all three masses when Grace+Lyra post; tau $\sqrt{\pi}$ residue forward heat-kernel derivation; address whether Szegő = 1 cancels in ratio (Cal #410 outcome b); offered direct run at Szegő kernel computation if Casey routes
Cal	#410+#411 banked; standing for FK reconciliation verdict (which outcome a/b/c); tier-tag at landing only; #387 α -under-own-lens still open; cold-read $d(\nu)$ explicit when posted
Keeper	This audit; self-correction #14 logged (K543 over-projection); Cal #408 standing maintained; honest projection +1 muon (not +2); tier-tag at landing; standing for FK reconciliation verdict

What's substantively right at K544

Substantive convergent close: - Lyra+Grace c_FK reconciliation framing (boundary vs bulk; π -structure distinction) - Cal #410 sharpened the question: 1.303 not 1; load-bearing not bookkeeping; two computable outcomes for investigation - Lyra honest framing: Szegő = 1 not automatic; compute don't assume - Grace explicit $d(\nu)$ polynomial delivered - Cal #411 projection correction: muon path +1; tau separate - Keeper self-correction #14: over-projection re-inflated tau

Cal #408 OPERATIONALIZED at meta-level: "seat correctly placed now: not gating; saying what tier; banking only what forward-derives and reconciles." Audit chain at peak.

Per Casey directive: investigate Szegő residue forward (Lyra) + verify $d(\nu)$ reproduces three masses (Elie) + tier-tag at landing (Cal). Count holds 4/26 disciplined. Honest projection: +1 (muon $\rightarrow$ 5) on FK reconciliation per Cal's two-outcome investigation; tau separate +1 only if forward residue clears its own look-elsewhere bar.

μ/τ asymmetry IS the content — Casey + Elie + Cal aligned on honest separation.

— Keeper, 2026-06-26 Friday substantive convergent close (365 cumulative discipline events; 14 Keeper self-corrections; Cal #410+#411 sharpens reconciliation question and projection; honest tier maintained; investigation unbounded + banking disciplined)